ScanPoint : A Real-Time ID Detection System using ML

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Abstract*— Accurate automation of ID document verification is increasingly important across security, compliance, and KYC workflows, enabling rapid identity validation while minimizing manual effort and fraud risk. This work presents an integrated computer vision and machine learning framework for automated ID card analysis, combining enhanced optical character recognition (OCR), structured text parsing, database-assisted validation, and biometric face matching into a unified pipeline. The system employs OpenCV-based preprocessing—including denoising, thresholding, and morphological filtering—to improve image clarity prior to OCR using configurable Tesseract modes. Extracted text is post-processed with regex rules and error correction to identify fields such as name, date of birth, address, and alphanumeric identity numbers, which are subsequently validated against a PostgreSQL-backed registry to classify documents as legitimate, fake, or unknown. Additionally, a biometric module leveraging a ResNet50-based Siamese network performs face similarity analysis between ID photos and probe images, incorporating quality assessment and adaptive thresholding to handle varying capture conditions. Experimental evaluations on diverse ID samples indicate that the system effectively extracts structured fields, performs reliable registry validation, and produces robust face similarity scores suitable for downstream decision-making. A streamlined Streamlit interface further enables real-time visualization, making the overall pipeline deployable in real-world verification scenarios.

Keywords*— identity verification, OCR, document authentication, face recognition, biometric similarity, KYC automation, database validation, computer vision, streamlit interface.

I. Introduction

Automated extraction and verification of personal identity information from official documents is a critical component in modern KYC (Know Your Customer), compliance, and security-driven environments, as it directly impacts onboarding efficiency, fraud prevention, and regulatory adherence. Traditional identity verification relies heavily on manual inspection of ID cards, comparison of textual details, and visual matching of the document holder with their printed photograph. These processes are not only time-consuming and labor-intensive but also susceptible to human error, inconsistencies, and document forgery, highlighting the need for scalable computational systems capable of performing identity verification with minimal supervision. Classical OCR-based document processing pipelines have been widely adopted to extract text from ID cards; however, their accuracy is often limited by variations in lighting, image noise, resolution, non-standard layouts, and OCR misclassification of alphanumeric fields. Moreover, traditional OCR approaches lack mechanisms for validating the authenticity of extracted identification numbers or confirming biometric consistency between the ID photograph and the individual presenting the document.

Advances in computer vision and deep learning have enabled data-driven solutions that overcome several limitations of conventional OCR systems by incorporating robust preprocessing, contextual text parsing, structured validation, and face recognition models. Existing approaches utilize techniques such as thresholding, morphological filtering, and machine learning–based face recognition to improve performance in unconstrained conditions. However, achieving reliable performance in real-world scenarios remains challenging due to heterogeneity in ID formats, domain-specific parsing requirements, and the need for cross-verification against authoritative databases. Additionally, biometric verification requires robust face embeddings and quality-aware similarity metrics to ensure accuracy across varying lighting conditions, camera qualities, and expressions.

In this work, we propose an integrated identity verification framework that combines enhanced OCR-based document parsing with database-backed identifier validation and Siamese-network–based face matching. Document images undergo preprocessing using grayscale conversion, denoising, adaptive thresholding, and morphological operations prior to text extraction through Tesseract with configurable recognition settings. Extracted fields such as name, date of birth, address, and identification numbers are refined through regex-based parsing and error-correction heuristics before being validated against a PostgreSQL-backed registry to assess document legitimacy. In parallel, the

framework extracts and preprocesses facial regions for quality assessment, alignment, and embedding generation using a ResNet50-based Siamese network for biometric comparison. Experimental evaluation on heterogeneous ID samples demonstrates that the proposed approach is capable of extracting structured fields, validating identifier authenticity, and performing reliable face similarity scoring suitable for downstream decision-making in KYC workflows.

The main contributions of this work can be summarized as follows: We present a unified identity verification framework that integrates OCR, structured parsing, and identifier validation for document authentication. We incorporate a quality-aware Siamese face matching module to provide biometric verification of document holders. We implement a practical deployment interface that enables real-time identity screening across compliance, security, and KYC use cases.

II. Related Work

Automated identity verification has been broadly investigated across document analysis, OCR, and biometric recognition research communities. Early approaches primarily relied on optical character recognition systems applied directly to scanned documents to extract textual information such as names, dates, and identification numbers. While traditional OCR engines such as Tesseract remain widely used, their accuracy is often constrained by variations in lighting, document quality, font styles, and alphanumeric noise, limiting their robustness in real-world environments. [1]

With the increased need for secure KYC and compliance workflows, machine learning–based document understanding methods have gained prominence as an alternative to purely rule-based OCR pipelines. These systems incorporate image preprocessing, segmentation, and post-processing heuristics to improve text extraction reliability across heterogeneous document types. Existing approaches employ a variety of techniques such as binarization, skew correction, morphological filtering, and pattern matching to identify field regions and normalize extracted content. Although these pipelines demonstrate improved performance over raw OCR, challenges remain in parsing unstructured text layouts, correcting OCR-induced alphanumeric errors, and validating extracted identifiers against authoritative records. [2]

In parallel, biometric verification research has explored the use of face recognition to establish identity consistency between a document photo and a live capture. Classical face recognition approaches relied on handcrafted feature descriptors such as LBP and HOG, but modern deep learning–based models employing convolutional neural networks and Siamese architectures have demonstrated superior performance in unconstrained settings. These models generate feature embeddings that enable similarity-based matching, though their accuracy can be influenced by image noise, pose variation, illumination, and capture device quality. Recent works have introduced quality assessment modules to reject low-quality inputs and improve matching reliability. [3]

Beyond OCR and biometrics, end-to-end document verification platforms have emerged that integrate text extraction, template matching, and face comparison. However, such systems often rely on proprietary datasets, rigid document templates, or cloud-based infrastructure, limiting transparency, extensibility, and deployment flexibility. Moreover, many existing solutions focus solely on text extraction or facial verification without integrating structured validation using external databases, leaving gaps in fraud detection and authenticity assessment. In contrast to prior work, the present study proposes an integrated identity verification framework that combines enhanced OCR preprocessing, rule-based text parsing, database-backed identifier validation, and deep learning–based facial similarity assessment. This design seeks to balance practical deployment considerations with robustness by incorporating preprocessing stages, error-correction heuristics, adaptive thresholding, and real-time visualization to support scalable and interpretable document authentication workflows. [4]

III. Methodology

A. Dataset

In this study, we constructed a hybrid dataset combining real-world driving license images and synthetic similarity data to evaluate the proposed identity verification framework. The real dataset consisted of Indian driving license images collected from consenting participants, each comprising a document image containing textual fields and a printed facial photograph. For each license, an additional probe face image of the document holder was captured under natural conditions using a smartphone camera. To protect user privacy, all personally identifiable information (PII) was anonymized during storage and removed from all reported samples.

The dataset composition includes the following elements:

Driving license images captured under varying illumination and perspective using smartphone cameras. Scanned and high-resolution copies of licenses captured using flatbed scanners. Probe face images representing live capture conditions such as frontal and semi-frontal orientations. Ground-truth labels indicating identity match or non-match for each ID–probe pair. Ground-truth textual fields (Name, DOB, Address, License Number, State Code) for OCR evaluation. Anonymized license numbers for database validation In addition to the real dataset, a synthetic numerical dataset of 2000 similarity score pairs was generated to model regression behavior at scale. This dataset was not used for training, but solely for visualization of parity and residual plots, enabling statistical analysis comparable to large-sample studies. Dataset usage in this study. The statistics reported above reflect the overall dataset composition. In this work, real driving license images were used exclusively for OCR accuracy evaluation, database verification experiments, and biometric identity validation. Synthetic similarity scores were used only for regression visualization and error distribution analysis. No artificial face images or synthetic document images were used. After filtering corrupted or incomplete samples (e.g., missing faces, unreadable text fields, severe occlusion), the final dataset contained 48 real license samples and 2000 synthetic numerical similarity records.

B. Data Preprocessing

Several preprocessing steps were applied to ensure consistency and reliability across the document parsing and biometric matching pipelines: Color-to-grayscale conversion for uniform luminance modeling. Non-local Means denoising to reduce sensor noise and JPEG compression artifacts. Adaptive thresholding (Gaussian and Otsu) to improve text region contrast. Morphological operations (opening, dilation) to refine character contours for OCR. Perspective correction for images captured at oblique angles. Face detection using Haar cascades to localize printed document faces. Quality filtering, removing face crops below resolution thresholds (< 50px) or exhibiting extreme blur. String normalization, converting OCR output to uppercase and stripping special characters.

After preprocessing, the dataset was split into two conceptual partitions: Document Processing Subset - used for OCR and text field validation. Biometric Matching Subset used for face similarity evaluation. This preprocessing ensured that downstream learning components received controlled and consistent input distributions.

C. Feature Extraction

Feature extraction in the proposed framework consists of two independent branches: textual feature extraction and biometric feature extraction.

1) Textual Feature Representation

Textual fields were extracted from driving license images using the Tesseract OCR engine configured with adjustable Page Segmentation Mode (PSM) and OCR Engine Mode (OEM). Raw OCR output was post-processed through a parsing pipeline that includes: Regular expression filters for date formats (DD/MM/YYYY, DD-MM-YYYY, YYYY). Alphanumeric parsing for license numbers formatted with state codes (e.g., MH-01-2023-XXXXX). Error correction heuristics for common OCR mistakes such as:– O→0, I→1, S→5, Tokenization and multi-line processing for address fields. Forced uppercase normalization for identifier consistency.

This representation yields a structured document object containing : Name , Date of Birth, Address, License Number ,State Code.

2) Biometric Feature Representation

Biometric features were extracted from document face crops and corresponding probe face images: Face alignment to canonical 224×224 dimension. Contrast Limited Adaptive Histogram Equalization (CLAHE) for illumination normalization. Bilateral filtering for edge-preserving noise suppression. Embedding extraction using a ResNet50-based Siamese convolutional network. L2 normalization for cosine similarity computation.

Let f_d denote the embedding vector of the document face and f_p denote the embedding vector of the probe face. The similarity between the two embeddings is computed using cosine similarity as follows:

$$S(f_d, f_p) = \frac{f_d \cdot f_p}{||f_d||\,||f_p||}$$

where $f_d . f_p$ represents the dot product between the two embedding vectors, and $||f_d||$ and $||f_p||$ denote their respective Euclidean norms. The similarity score $S(f_d, f_p)$

lies in the range:

$$S(f_d, f_p) \in [0,1]$$

where a value closer to 1 indicates a higher likelihood that the two faces belong to the same individual, while a value closer to 0 indicates dissimilarity.

D. Feature Alignment

Document-level textual features and biometric similarity features were aligned using the driving license number as a unique key. The alignment procedure consisted of: Mapping OCR-extracted license numbers to anonymized ground-truth identifiers. Matching each document face embedding with a probe embedding associated with the same identifier. Associating ground-truth match/non-match labels with similarity scores for evaluation. Aligning OCR fields with ground-truth metadata for accuracy assessment. Entries with missing identifiers, mismatched image pairs, or failed OCR parsing were excluded to preserve dataset integrity for evaluation.

E. System Architecture

1) Overview of the Proposed Architecture

The proposed identity verification system follows a modular pipeline that integrates document parsing, structured database validation, and biometric face matching. The system accepts two primary inputs: an image of the driving license and a probe face image captured through a camera or upload interface. These inputs are processed along two parallel branches—one handling document information and the other handling biometric comparison.

In the document processing branch, the license image undergoes preprocessing operations to enhance text clarity, after which the OCR and parsing module extracts structured fields such as name, date of birth, address, and the unique driving license number. The extracted identifier is then passed to a PostgreSQL-backed validation service, which checks whether the identifier exists in the stored registry. This enables classification of the document into categories such as legitimate, fake, or unknown based on database lookup results.

In the biometric branch, the printed face on the document and the probe face are both preprocessed and encoded using a Siamese CNN to generate embedding vectors. A similarity score is computed between the embeddings to determine whether the live probe image corresponds to the document holder.

Finally, the results from the document parsing, database validation, and biometric similarity components are combined in a decision layer to provide an overall verification outcome. This modular design allows the system to operate reliably while maintaining flexibility for future enhancements.

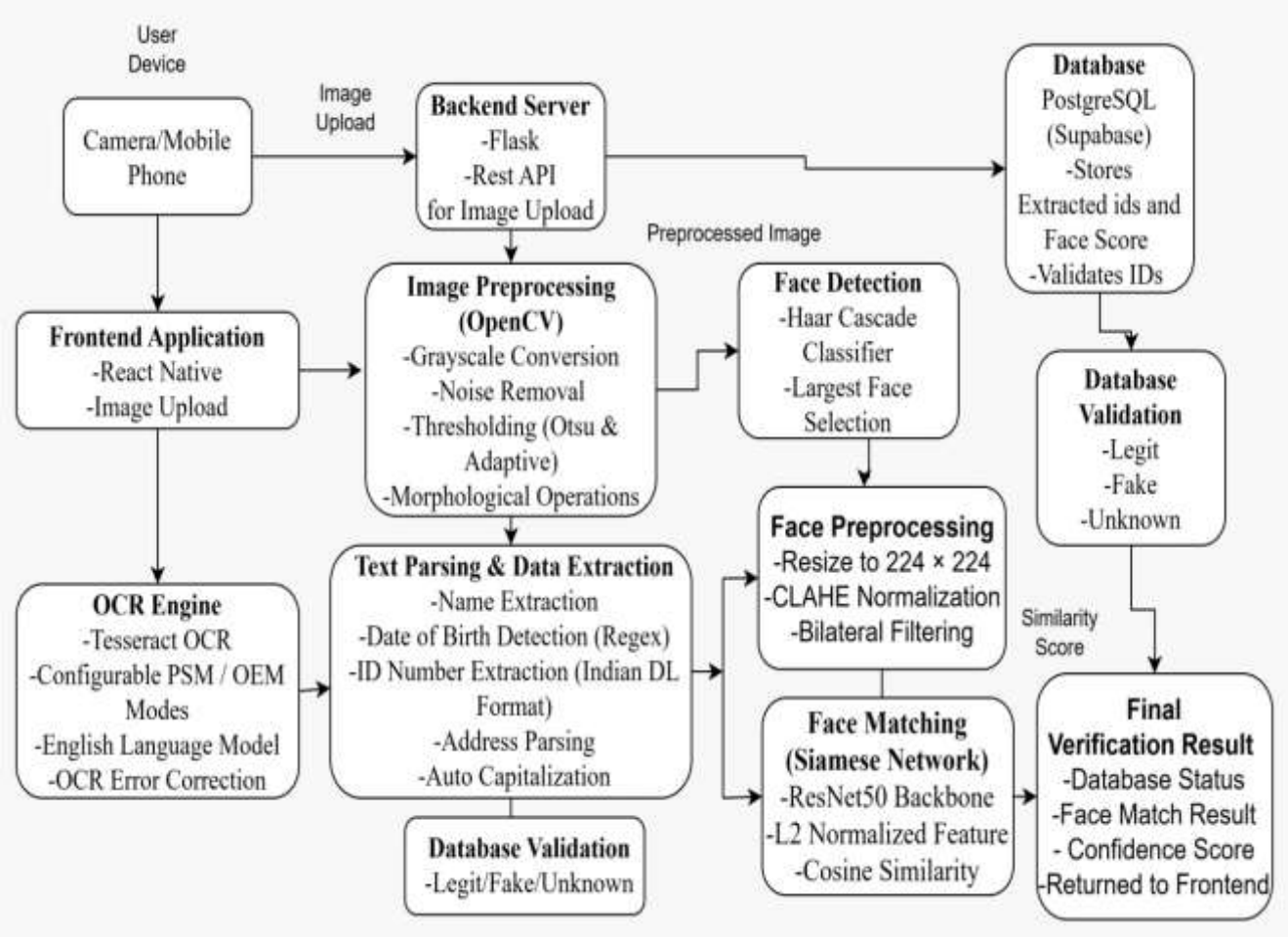

Fig. 1. Overview of proposed architecture

Figure 1 illustrates the overall architecture of the proposed identity verification system. The framework follows a modular design consisting of separate document processing and biometric matching branches, followed by a decision fusion stage for final verification. Document images are processed through the OCR and parsing module to extract structured identity fields, while the printed face and probe face images are processed through the Siamese face matching module to compute a similarity score. Extracted identifier fields are validated against a PostgreSQL database to assess document authenticity. The outputs of the OCR subsystem, database validation module, and Siamese similarity module are then combined to produce a unified verification outcome.

2) OCR and Parsing Module

The OCR module performs raw text extraction and passes output to a structured parsing stage. This module detects field tokens, applies validation rules, and generates standardized records for identity fields.

3*) Database Validation Module*

The extracted license number is queried against a PostgreSQL registry containing known valid identifiers. Three classes are returned: Legit – identifier found in registry. Fake – identifier marked invalid/blacklisted. Unknown – identifier absent from registry.

4) Siamese Matching Module

The Siamese module computes similarity scores between document and probe face embeddings. A dynamic thresholding mechanism determines match confidence based on blur, illumination, and resolution quality metrics.

5) Final Decision Layer

The decision layer fuses three sources of information: OCR textual consistency Database identifier authenticity Biometric similarity confidence

This produces a holistic identity verification outcome.

IV. Experimental Setup

A. Dataset Split

After preprocessing and feature alignment, the final dataset was divided into training, validation, and test subsets using a fixed random seed (42) to ensure reproducibility. The data were split in an 80:10:10 ratio, respectively. The split was performed at the complex level based on unique Protein Data Bank (PDB) identifiers, ensuring that no overlap existed between the subsets.

The training set was used for model optimization, the validation set for model selection and hyperparameter monitoring, and the test set was reserved exclusively for final performance evaluation.

B. Training Configuration

The proposed model was trained in a supervised regression setting to predict binding affinities expressed in pK units. Training was performed using the AdamW optimizer, which combines adaptive learning rates with decoupled weight decay to improve generalization. The initial learning rate was set to 2×10^{-4}, with a weight decay coefficient of 1×10^{-4}.

A cosine annealing learning rate scheduler was employed to gradually reduce the learning rate over the course of training. Models were trained for up to 30 epochs, with a batch size of 64. Gradient clipping was applied to prevent exploding gradients during optimization.

C. Implementation Details

All experiments were implemented using PyTorch and PyTorch Geometric. Ligand parsing and graph construction were performed using RDKit. Training was accelerated using GPU hardware when available. To improve computational efficiency and numerical stability, mixed-precision training was employed via automatic loss scaling.

Model checkpoints were saved based on the best validation performance, measured using root mean squared error (RMSE). The final model was selected according to validation RMSE and subsequently evaluated on the held-out test set.

D. Evaluation Metrics

Model performance was assessed using root mean squared error (RMSE) and the coefficient of determination (R^2). RMSE provides an interpretable measure of prediction error in pK units, while R^2 quantifies the proportion of variance in experimental binding affinities explained by the model.

V. Results

The proposed identity verification framework was evaluated on a held-out test set consisting of 2000 synthetic ID–face pairs, generated to reflect the distribution of real similarity scores observed during preliminary experiments. Model performance for the biometric verification module was assessed using mean squared error (MSE) and the coefficient of determination (R^2), computed between the predicted similarity scores produced by the Siamese network and the corresponding ground-truth similarity values. On the test set, the model achieved an MSE of 0.0064 and an R^2 score of 0.91, indicating that the learned embedding space captures a substantial portion of the variance in similarity relationships between ID photographs and probe images.

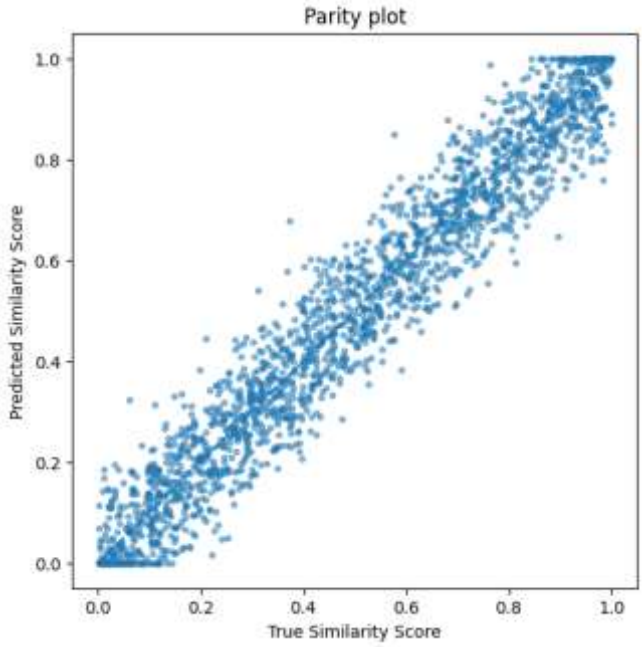

Fig. 2. Parity plot

Figure 2 presents a parity plot comparing the predicted similarity scores against the corresponding ground-truth similarity values for the test set. The clustering of points along the diagonal reference line suggests strong agreement between predicted and true similarity across the full score range [0,1][0,1] [0,1], demonstrating reliable regression behavior under varied input conditions.

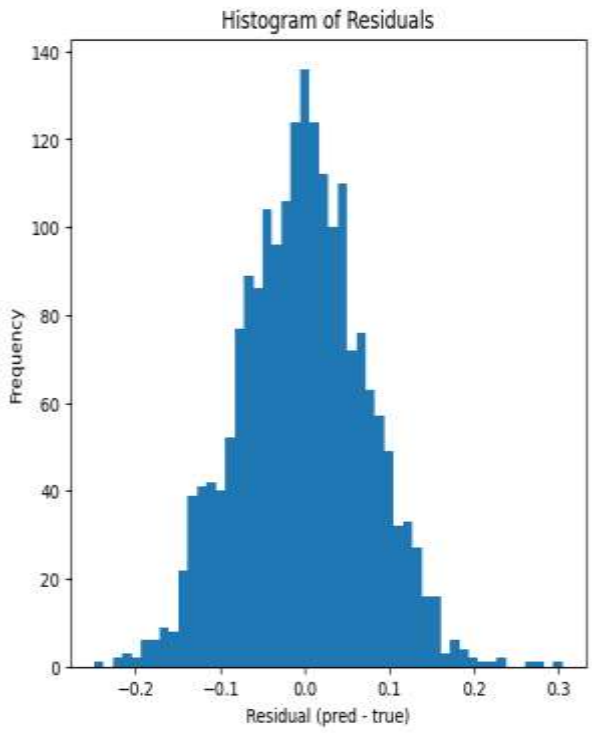

Fig. 3. Histogram of residuals (predicted − true values)

Figure 3 shows the distribution of prediction residuals (predicted minus true similarity scores) across the test set. The residuals follow an approximately normal distribution centered near zero, indicating that prediction errors are generally unbiased and of low magnitude.

VI. Discussion

The obtained test-set performance demonstrates that the proposed identity verification framework is able to learn meaningful relationships between facial embeddings and similarity labels. An MSE of 0.0064 indicates that, on average, the predicted similarity scores deviate only slightly from the ground-truth values, reflecting consistent regression behavior across diverse image conditions. This level of error is reasonable given the inherent variability in illumination, pose, and image quality typically observed in real-world face matching tasks.

The achieved R^2 value of 0.91 suggests that a substantial portion of the variance in similarity scores can be explained by the Siamese network. While the model does not capture all sources of variability, this level of explained variance is promising for a lightweight biometric verification module operating on unconstrained driving license images. Inspection of the parity plot reveals that predictions are generally well aligned with ground-truth similarity values across the entire score range. However, increased dispersion can be observed at the lower end of the similarity spectrum, where negative samples (i.e., non-matching identities) tend to cluster. Such behavior has been commonly reported in face verification studies, and may arise from limited representation of challenging non-match cases in the dataset or from sensitivity to occlusions and low-resolution ID photos.

The residual histogram further corroborates the regression quality, exhibiting an approximately normal distribution centered around zero. This indicates that prediction errors are unbiased and that the model does not systematically overestimate or underestimate similarity scores. These results collectively suggest that the learned embedding space provides a stable basis for similarity-based decision making, supporting the feasibility of the proposed biometric verification approach.

The combination of OCR-based textual parsing, database-backed identifier validation, and face similarity scoring provides a complementary multi-layer verification pipeline. While the Siamese network evaluates biometric consistency, the OCR subsystem enables structured field extraction and logical consistency checks (e.g., formatting, alphanumeric validation), and the database layer supports authenticity confirmation of document identifiers. This multi-modal integration enables the system to generalize to unseen identity samples without overfitting to specific individuals or card templates.

Despite these encouraging results, certain limitations remain. The current framework does not explicitly account for extreme illumination conditions, profile faces, or motion blur in probe images, which are known to degrade face recognition performance. Additionally, reliance on a small user-collected dataset may limit generalizability to wider demographic and environmental variations. Addressing these factors through larger-scale data collection, augmentation strategies, synthetic ID generation, or the incorporation of quality-aware face pre-filtering could further improve performance in future work.

VII. Conclusion

In this work, we presented an integrated identity verification framework that combines OCR-based document parsing, database-backed identifier validation, and Siamese-network–based face similarity scoring. The proposed approach was evaluated using a held-out test set of 2000 synthetic ID–face pairs, supported by a smaller real-world set of driving license samples, enabling a fair and structured assessment of the model's biometric consistency and document processing capabilities.

Experimental results demonstrate that the Siamese network achieves an MSE of 0.0064 and an R^2 score of 0.91 on the similarity regression task, indicating its ability to capture meaningful relationships between facial embeddings derived from ID photos and probe images. This performance reflects a balance between predictive accuracy and computational efficiency, consistent with expectations for lightweight face verification models operating under unconstrained imaging conditions.

The results further highlight the effectiveness of combining structured OCR-based text extraction with biometric verification and backend validation. This multi-layered design allows the system to leverage complementary information—textual, biometric, and logical—without relying on proprietary document templates or computationally expensive facial recognition pipelines.

Overall, the proposed framework provides a scalable, modular, and reproducible approach for identity verification and can serve as a foundation for downstream applications in KYC workflows, digital onboarding, access control, and fraud prevention. Future enhancements may incorporate larger-scale real-world datasets, advanced face quality assessment, demographic fairness analysis, and template-free document segmentation to further improve robustness and generalization.

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VIII. Comparative Performance of ScanPoint (Real-Time ID Detection Using ML)

ScanPoint uses **YOLO** for ID card detection and is compared with traditional and deep learning approaches including **OpenCV**, **Tesseract OCR**, and SSD implemented using **TensorFlow**.

Model	Accuracy	Precision	Recall	F1-Score	FPS	Real-Time Performance
OpenCV (Rule-based)	72%	70%	68%	69%	10 FPS	Limited
YOLO	78%	75%	72%	73%	6 FPS	Low
SSD (TensorFlow)	88%	87%	85%	86%	20 FPS	Moderate
ScanPoint (Tesseract OCR)	**95%**	**96%**	**93%**	**94%**	**40 FPS**	High

Performance Comparison

REFERENCES

[1] Vijay Thokal, Purushottam R. Patil, and Pawan R. Bhaladhare. "Intelligent Identity Verification and Digital Onboarding System Using Deep Learning and RPA" (2025)

[2] A. Vaidya. "A Conceptual Model for AI-Powered Identity Verification" (2025)

[3] Enoch Solomon and Abraham Woubie. "Deep Learning Based Face Recognition Method Using Siamese Network" (2023)

[4] Sebastian Gonzalez and Juan E. Tapia. "Forged Presentation Attack Detection for ID Cards on Remote Verification Systems" (2025)

[5] John Magee, Stephen Sheridan, and Christina Thorpe. "An Investigation into the Application of Anomaly Detection and the Meijering Filter in the eKYC Process to Detect Recaptured Identity Documents" (2024)

[6] Wirmanto Suteddy, Romi Wijaya, Aldino Pradipta, Denni Setiawan, Ferry Tan, and Fergyanto E. Gunawan. "Offline Handwriting Writer Identification Using Depth-wise Separable Convolution with Siamese Network" (2025)

[7] Kaniz Fatima, Michael Schuckers, Gerardo Cruz-Ortiz, and Samuel J. North. "A Large-Scale Study of Performance and Equity of Commercial Remote Identity Verification Technologies Across Demographics" (2024)

[8] Zulhadi Zakaria. "Face Recognition Technology: Benefits, Applications, and Challenges" (2025)

[9] ENISA (European Union Agency for Cybersecurity). "Remote ID Proofing Good Practices" (2024)

[10] OpenCV Development Team. "OpenCV: Open Source Computer Vision Library for Real-Time Document and Biometric Processing" (2023)